

OpenEXR Attributes Source Extension for RDD18, Sony Private Acquisition Metadata, and Cooke Protocol Metadata

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1. General

1.1 Scope

This document defines extensions to the following metadata, based on the [OpenEXR Attributes Base Specification for Camera Acquisition Metadata](#).

- [SMPTE RDD 18:2021 Acquisition Metadata Sets](#) (<https://pub.smpte.org/doc/rdd18/20210701-pub/>)
- [Sony Private Acquisition Metadata Set Version 1.10](#), which is a Sony-specific extension to the [SMPTE RDD 18](#). This also defines a metadata set for storing Lens Metadata compliant with the [Cooke /i Protocol](#) and its extensions.

1.2 Source Metadata

This specification primarily covers the following information to be stored in OpenEXR Header Attributes:

- Metadata Sets defined in [SMPTE RDD 18:2021 Acquisition Metadata Sets](#) specification.
- Metadata Sets defined in [Sony Private Acquisition Metadata Set Version 1.10](#) specification. These Metadata Sets are Sony-specific extensions implemented using the User Defined Acquisition Metadata Set mechanism for extending the above [SMPTE RDD 18](#) specification.
- Metadata Set for lens metadata compliant with [Cooke /i Protocol](#) and its extensions. This is defined as one of the Metadata Sets of the above [Sony Private Acquisition Metadata Set](#) specification.

All of the above metadata sets are recorded as Data Items in the Essence Container of an MXF file. In addition, these metadata sets MAY also be embedded in an SDI signal as ancillary data packets.

1.3 OpenEXR Header Attributes

1.3.1 OpenEXR Specification

The OpenEXR Header Attributes defined in this specification SHALL conform to the [OpenEXR Attributes Base Specification for Camera Acquisition Metadata](#), which further conforms to the following specification:

- OpenEXR Specification, version 3.x (<https://openexr.com/en/latest/index.html>)

Equivalent specifications are also defined in the following:

- SMPTE ST 2065-4:2023, ACES Image Container File Layout
-

2. Convention Policy

This section describes the policy for defining extensions to OpenEXR attributes.

2.1 Base Policy

The policy of the Base Specification ([OpenEXR Attributes Base Specification for Camera Acquisition Metadata](#)) SHALL be followed.

2.1 Namespace

All OpenEXR attributes SHALL have a namespace identifier beginning with `acq:` . The Base Specification further defines the following second-level sub-namespaces:

- MXF sub-namespace (`mxf:`)
- Lens sub-namespace (`lens:`)
- Camera sub-namespace (`camera:`)
- Imager sub-namespace (`imager:`)
- Color sub-namespace (`color:`)
- Slate sub-namespace (`slate:`)

In addition to above, this specification defines the following second-level sub-namespaces:

- RDD 18 sub-namespace (`rdd18:`)
- Sony Private sub-namespace (`sony:`)
- Cooke sub-namespace (`cooke:`)

2.2 Preference for Unified Attributes

For OpenEXR attributes in the sub-namespaces extended by this document, attributes that represent common concepts SHOULD be normalized and stored as vendor-neutral attributes defined by the Base Specification. In such cases, specification-specific or vendor-specific attributes belonging to the sub-namespaces extended by this document MAY also be stored together with the normalized attributes.

If there is a concern that information may be lost by normalization into vendor-neutral attributes, specification-specific or vendor-specific information SHOULD be referenced as necessary.

Furthermore, metadata that can be represented using OpenEXR Standard Attributes SHOULD be expressed using those attributes. In such cases, vendor-neutral attributes defined by this specification and specification/vendor-specific attributes MAY also be stored together with the OpenEXR Standard Attributes.

3 Metadata Mapping

3.1 Overview

The specification-specific or vendor-specific attributes defined in this specification use the namespaces shown in the table below.

Namespace Identifier	Description for the source information
<code>acq:rdd18:</code>	SMPTE RDD 18 specification-specific metadata
<code>acq:sony:</code>	Sony Private Acquisition Metadata Set specification-specific metadata
<code>acq:cooke:</code>	Coole i/ Protocol specification-specific metadata

Additional Sub-namespace Identifiers may be appended after the Namespace Identifier described above, as needed.

3.2 Schema Identifier Attributes

3.2.1 SMPTE RDD 18 Schema Identifier Attributes

The following schema identifier SHALL be used to describe attributes in the RDD 18 namespace.

Attribute Name	Type	Requirement	Value
<code>acq:rdd18:schema:standard</code>	<code>string</code>	REQUIRED when <code>rdd18:*</code> is present	"SMPTE RDD 18"
<code>acq:rdd18:schema:version</code>	<code>string</code>	REQUIRED when <code>rdd18:*</code> is present	"2021"

3.2.2 Sony Private Acquisition Metadata Schema Identifier Attributes

The following schema identifier SHALL be used to describe attributes in the Sony Private Acquisition Metadata namespace.

Attribute Name	Type	Requirement	Value
acq:sony:schema:standard	string	REQUIRED when <code>sony:*</code> is present	"Sony Private Acquisition Metadata"
acq:sony:schema:version	string	REQUIRED when <code>sony:*</code> is present	"1.10"

3.2.3 Cooke Protocol Schema Identifier Attributes

The following schema identifier SHALL be used to describe attributes in the Cooke Protocol namespace.

Attribute Name	Type	Requirement	Value
acq:cooke:schema:standard	string	REQUIRED when <code>cooke:*</code> is present	"Cooke i/ Protocol"
acq:cooke:schema:version	string	REQUIRED when <code>cooke:*</code> is present	"5.0"

3.3 RDD 18 Attributes

3.3.1 Lens Unit Metadata Set

3.3.1.1 RDD 18 Native Notation

When storing information obtained from each item of the RDD 18 Lens Unit Metadata Set as RDD 18-specific OpenEXR Attributes, the definitions in the table below SHALL be followed.

However, where normalization to vendor-neutral Lens Attributes defined in the Base Specification is possible, they SHOULD be normalized in accordance with Section 3.3.1.2.

Also, where normalization to OpenEXR Standard Attributes is possible, they SHOULD be normalized in accordance with Section 3.3.1.3.

RDD 18 Item Name	RDD 18-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
Iris F-Number	acq:rdd18:lens:irisFNumber	unsigned short
Iris T-Number	acq:rdd18:lens:irisTNumber	unsigned short
Iris Ring Position	acq:rdd18:lens:irisRingPosition	unsigned short
Focus Position From Image Plane	acq:rdd18:lens:focusPositionFromImagePlane	float
Focus Position From Front Lens Vertex	acq:rdd18:lens:focusPositionFromFrontLensVertex	float
Focus Ring Position	acq:rdd18:lens:focusRingPosition	unsigned short
Macro Setting	acq:rdd18:lens:macroSetting	unsigned char
Lens Zoom 35mm Still Camera Equivalent	acq:rdd18:lens:lensZoom35mmStill Camera Equivalent	float
Lens Zoom Actual Focal Length	acq:rdd18:lens:lensZoomActualFocallLength	float
Zoom Ring Position	acq:rdd18:lens:zoomRingPosition	unsigned short
Anamorphic Lens Squeeze Ratio	acq:rdd18:lens:anamorphicLensSqueezeRatio	unsigned char
Optical Inversion	acq:rdd18:lens:opticalInversion	unsigned char
OpticalExtender Magnification	acq:rdd18:lens:opticalExtenderMagnification	unsigned short
Lens Attributes	acq:rdd18:lens:lenAttributes	string

3.3.1.2 Normalization to Vendor-Neutral Lens Attributes

Among the items in the RDD 18 Lens Unit Metadata Set, those listed in the table below can be recorded as the corresponding vendor-neutral Lens Attributes, and therefore these SHOULD take precedence.

RDD 18 Item Name	Vendor-Neutral OpenEXR Attribute Name	OpenEXR Attribute Type	Conversion
Iris F-Number	<code>acq:lens:irisFNumber</code>	float	Decode RDD18 iris encoding to F-number
Iris T-Number	<code>acq:lens:irisTNumber</code>	float	Decode to T-stop
Focus Position From Image Plane	<code>acq:lens:focusDistance</code>	float	meters to millimeters
Focus Position From Front Lens Vertex	<code>acq:lens:focusDistance</code>	float	fallback only; meters to millimeters
Lens Zoom Actual Focal Length	<code>acq:lens:effectiveFocalLength</code>	float	meters to millimeters
Optical Inversion	<code>acq:lens:opticalInversion</code>	int	Preserve code
Optical Extender Magnification	<code>acq:lens:opticalExtenderMagnification</code>	int	percent
Lens Attributes	<code>acq:lens:model</code>	string	

3.3.1.3 Normalization to OpenEXR Standard Attributes

Among the items in the RDD 18 Lens Unit Metadata Set, those listed in the table below can be stored as the corresponding OpenEXR Standard Attributes, and therefore these SHOULD take precedence.

RDD 18 Item Name	OpenEXR Standard Attribute Name	OpenEXR Attribute Type
Iris F-Number	<code>aperture</code>	float
Iris T-Number	<code>tStop</code>	float
Focus Position From Image Plane	<code>focus</code>	float
Focus Position From Front Lens Vertex	<code>focus</code>	float
Image Sensor Dimension Effective Width Image Sensor Dimension Effective Height	<code>sensorAcquisitionRectangle</code>	box2i
Shutter Speed (Time)	<code>expTime</code>	rational
ISO Sensitivity ISO Sensitivity in Long Integer	<code>isoSpeed</code>	float
White Balance	<code>cameraCCTSetting</code>	int
Tint Correction	<code>cameraTintSetting</code>	float

3.3.2 Camera Unit Metadata Set

3.3.2.1 RDD 18 Native Notation

When storing information obtained from each item of the RDD 18 Camera Unit Metadata Set as RDD 18-specific OpenEXR Attributes, the definitions in the table below SHALL be followed.

However, where normalization to vendor-neutral Camera Attributes defined in the Base Specification is possible, they SHOULD be normalized in accordance with Section 3.3.2.2.

Also, where normalization to OpenEXR Standard Attributes is possible, they SHOULD be normalized in accordance with Section 3.3.2.3.

RDD 18 Item Name	RDD 18-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
Auto Exposure Mode	<code>acq:rdd18:camera:autoExposureMode</code>	string
Exposure Index of Photo Meter	<code>acq:rdd18:camera:exposureIndexofPhotoMeter</code>	unsigned short
Exposure Index of Photo Meter in Long Integer	<code>acq:rdd18:camera:exposureIndexofPhotoMeterInLong Integer</code>	unsigned int
Auto Focus Sensing Area Setting	<code>acq:rdd18:camera:autoFocusSensingAreaSetting</code>	unsigned char
Color Correction Filter Wheel Setting	<code>acq:rdd18:camera:colorCorrectionFilterWheelSetting</code>	unsigned char
Neutral Density Filter Wheel Setting	<code>acq:rdd18:camera:neutralDensityFilterWheelSetting</code>	unsigned short
Rotary Shutter	<code>acq:rdd18:camera:Rotary Shutter</code>	unsigned char
Optical Anti-Aliasing Filter	<code>acq:rdd18:camera:opticalAntiAliasingFilter</code>	unsigned char
Image Sensor Dimension Effective Width	<code>acq:rdd18:camera:imageSensorDimensionEffectiveWidth</code>	unsigned short
Image Sensor Dimension Effective Height	<code>acq:rdd18:camera:imageSensorDimensionEffectiveHeight</code>	unsigned short
Capture Frame Rate	<code>acq:rdd18:camera:captureFrameRate</code>	rational
Image Sensor Readout Mode	<code>acq:rdd18:camera:imageSensorReadoutMode</code>	unsigned char
Image Sensor Decimation Ratio	<code>acq:rdd18:camera:imageSensorDecimationRatio</code>	unsigned char
Image Scan Direction	<code>acq:rdd18:camera:imageScanDirection</code>	unsigned char
Shutter Speed (Angle)	<code>acq:rdd18:camera:shutterSpeedAngle</code>	unsigned int
Shutter Speed (Time)	<code>acq:rdd18:camera:shutterSpeedTime</code>	rational
Camera Master Gain Adjustment	<code>acq:rdd18:camera:cameraMasterGainAdjustment</code>	short
ISO Sensitivity	<code>acq:rdd18:camera:isoSensitivity</code>	unsigned short
ISO Sensitivity in Long Integer	<code>acq:rdd18:camera:isoSensitivityInLongInteger</code>	unsigned int
Color Matrix	<code>acq:rdd18:camera:colorMatrix:rToR</code>	rational
	<code>acq:rdd18:camera:colorMatrix:gToR</code>	rational
	<code>acq:rdd18:camera:colorMatrix:bToR</code>	rational
	<code>acq:rdd18:camera:colorMatrix:rToG</code>	rational
	<code>acq:rdd18:camera:colorMatrix:gToG</code>	rational
	<code>acq:rdd18:camera:colorMatrix:bToG</code>	rational
	<code>acq:rdd18:camera:colorMatrix:rToB</code>	rational
	<code>acq:rdd18:camera:colorMatrix:gToB</code>	rational
	<code>acq:rdd18:camera:colorMatrix:bToB</code>	rational
Electrical Extender Magnification	<code>acq:rdd18:camera:electricalExtenderMagnification</code>	unsigned short
Auto White Balance Mode	<code>acq:rdd18:camera:autoWhiteBalanceMode</code>	unsigned char
White Balance	<code>acq:rdd18:camera:whiteBalance</code>	unsigned short
Tint Correction	<code>acq:rdd18:camera:tintCorrection</code>	short
Camera Master Black Level	<code>acq:rdd18:camera:cameraMasterBlackLevel</code>	short
Camera Knee Point	<code>acq:rdd18:camera:cameraKneePoint</code>	unsigned short

RDD 18 Item Name	RDD 18-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
Camera Knee Slope	<code>acq:rdd18:camera:cameraKneeSlope</code>	rational
Camera Luminance Dynamic Range	<code>acq:rdd18:camera:cameraLuminanceDynamicRange</code>	unsigned short
Capture Gamma Equation	<code>acq:rdd18:camera:captureGammaEquation</code>	string
Capture Color Primaries	<code>acq:rdd18:camera:captureColorPrimaries</code>	string
Coding Equations	<code>acq:rdd18:camera:codingEquations</code>	string
Luminance Code Range	<code>acq:rdd18:camera:luminanceCodeRange</code>	unsigned char
Gamma for CDL	<code>acq:rdd18:camera:gammaForCd1</code>	unsigned char
Color for CDL	<code>acq:rdd18:camera:colorForCd1</code>	unsigned char
ASC CDL V1.2	<code>acq:rdd18:camera:ascCd1V1_2:slope</code>	v3f
	<code>acq:rdd18:camera:ascCd1V1_2:offset</code>	v3f
	<code>acq:rdd18:camera:ascCd1V1_2:power</code>	v3f
	<code>acq:rdd18:camera:ascCd1V1_2:saturation</code>	float
Acquisition Setting Procedure	<code>acq:rdd18:camera:acquisitionSettingProcedure</code>	string
Camera Setting File URI	<code>acq:rdd18:camera:cameraSettingFileUri</code>	string
Camera Attributes	<code>acq:rdd18:camera:cameraAttributes</code>	string

3.3.2.2 Normalization to Vendor-Neutral Camera Attributes

Among the items in the RDD 18 Camera Unit Metadata Set, those listed in the table below can be stored as the corresponding vendor-neutral Lens Attributes, and therefore these SHOULD take precedence.

RDD 18 Item Name	Vendor-Neutral OpenEXR Attribute Name	OpenEXR Attribute Type	Conversion
Exposure Index of Photo Meter Exposure Index of Photo Meter in Long Integer	<code>acq:camera:exposureIndex</code>	float	
Neutral Density Filter Wheel Setting	<code>acq:camera:ndFilterIndex</code>	int	
Image Sensor Dimension Effective Width Image Sensor Dimension Effective Height	<code>acq:capture:ndFilterIndex</code>	v2f	
Capture Frame Rate	<code>Iacq:camera:captureFrameRate</code>	rational	
Shutter Speed (Time)	<code>acq:camera:exposureTime</code>	rational	
ISO Sensitivity ISO Sensitivity in Long Integer	<code>acq:camera:isoSensitivity</code>	float	
White Balance	<code>acq:color:colorTemperature</code>	int	
Tint Correction	<code>acq:color:tint</code>	float	
Capture Gamma Equation	<code>acq:color:captureGammaEquation</code>	string	
Capture Color Primaries	<code>acq:color:captureColorPrimaries</code>	string	
Coding Equations	<code>acq:color:codingEquations</code>	string	
Luminance Code Range	<code>acq:color:luminanceCodeRange</code>	int	

RDD 18 Item Name	Vendor-Neutral OpenEXR Attribute Name	OpenEXR Attribute Type	Conversion
ASC CDL V1.2	<code>acq:camera:ascCd1:slope</code>	v3f	
	<code>acq:camera:ascCd1:offset</code>	v3f	
	<code>acq:camera:ascCd1:power</code>	v3f	
	<code>acq:camera:ascCd1:saturation</code>	float	
Camera Attributes	<code>acq:camera:model</code>	string	

3.3.2.3 Normalization to OpenEXR Standard Attributes

Among the items in the RDD 18 Camera Unit Metadata Set, those listed in the table below can be stored as the corresponding OpenEXR Standard Attributes, and therefore these SHOULD take precedence.

RDD 18 Item Name	OpenEXR Standard Attribute Name	OpenEXR Attribute Type
Image Sensor Dimension Effective Width Image Sensor Dimension Effective Height	<code>sensorAcquisitionRectangle</code>	box2i
Shutter Speed (Time)	<code>expTime</code>	rational
ISO Sensitivity ISO Sensitivity in Long Integer	<code>isoSpeed</code>	float
White Balance	<code>cameraCCTSetting</code>	int
Tint Correction	<code>cameraTintSetting</code>	float

3.4 Sony Private Acquisition Attributes

3.4.1 Cooke Protocol Lens Metadata Set

Note that this section does not describe the Cooke Protocol metadata itself, but rather the items in the Sony Private Acquisition Metadata Set that carry the Cooke Protocol metadata.

When storing information obtained from each item of the Cooke Protocol Lens Metadata Set as Sony-specific OpenEXR Attributes, the definitions in the table below SHALL be followed. However, where normalization to vendor-neutral Attributes defined in the Base Specification is possible, or OpenEXR Standard Attributes, they SHOULD be stored as such.

Note that, in the Sony Private Acquisition Metadata Set, the Cooke Protocol lens data is divided and carried in separate Metadata Items. Therefore, it is preferable that the Translator either reassembles the divided packets to reconstruct the original Cooke Protocol lens data string, or further parses it into individual attributes as described in Section 3.5.

Cooke Protocol Lens Metadata Set Item Name	Sony-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
UDAM Set Identifier	<code>acq:sony:cooke:udamSetIdentifier</code>	string
Cooke Protocol Start-up Command Leader Part	<code>acq:sony:cooke:startupCommandLeaderPart</code>	string
Cooke Protocol Start-up Command Shading Part	<code>acq:sony:cooke:cookeProtocolStartupCommandShadingPart</code>	string
Cooke Protocol Start-up Command Distortion Part	<code>acq:sony:cooke:startupCommandDistortionPart</code>	string
Cooke Protocol Start-up Command Inertial Cal Part	<code>acq:sony:cooke:startupCommandInertialCalPart</code>	string
Cooke Protocol Start-up Command Trailer Part	<code>acq:sony:cooke:startupCommandTrailerPart</code>	string
Cooke Protocol Start-up Command Extension 1	<code>acq:sony:cooke:startupCommandExtension1</code>	string
Cooke Protocol Start-up Command Extension 2	<code>acq:sony:cooke:startupCommandExtension2</code>	string

Cooke Protocol Lens Metadata Set Item Name	Sony-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
Cooke Protocol Start-up Command Extension 3	<code>acq:sony:cooke:startupCommandExtension3</code>	string
Cooke Protocol Start-up Command Extension 4	<code>acq:sony:cooke:startupCommandExtension4</code>	string
Cooke Protocol Calibration Type	<code>acq:sony:cooke:calibrationType</code>	string
Cooke Protocol Binary Metadata	<code>acq:sony:cooke:binaryMetadata</code>	string
Cooke Protocol Inertial Tracking Data	<code>acq:sony:cooke:inertialTrackingData</code>	string
Cooke Protocol Inertial Tracking Data Extension	<code>acq:sony:cooke:inertialTrackingData Extension</code>	string
Cooke Protocol Carl Zeiss AG Dataset	<code>acq:sony:cooke:carlZeissAgDataset</code>	string
Cooke Protocol Carl Zeiss AG Advanced Dataset	<code>acq:sony:cooke:carlZeissAgAdvancedDataset</code>	string
Cooke Protocol User Metadata	<code>acq:sony:cooke:userMetadata</code>	string
Lens Attribute	<code>acq:sony:cooke:lensAttribute</code>	string

3.4.2 Sony F65 Camera Metadata Set

3.4.2.1 Sony Private Acquisition Metadata Native Notation

When storing information obtained from each item of the Sony F65 Camera Metadata Set as Sony-specific OpenEXR Attributes, the definitions in the table below SHALL be followed.

However, where normalization to vendor-neutral Attributes defined in the Base Specification is possible, or OpenEXR Standard Attributes, they SHOULD be stored as such.

Sony F65 Camera Metadata Set Item Name	Sony-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
UDAM Set Identifier	<code>acq:sony:f65:udamSetIdentifier</code>	string
Effective Marker Coverage	<code>acq:sony:f65:effectiveMarkerCoverage</code>	rational
Effective Marker Aspect Ratio	<code>acq:sony:f65:Effective Marker Aspect Ratio</code>	rational
User Frame Line 1	<code>acq:sony:f65:userFrameLine1</code>	v2i
User Frame Line 2	<code>acq:sony:f65:userFrameLine2</code>	v2i
User Frame Line 3	<code>acq:sony:f65:UserFrameLine3</code>	v2i
User Frame Line 4	<code>acq:sony:f65:UserFrameLine4</code>	v2i
Active Area Aspect Ratio	<code>acq:sony:f65:activeAreaAspectRatio</code>	rational
Pixel Aspect Ratio	<code>acq:sony:f65:pixelAspectRatio</code>	rational
Camera Process Discrimination Code	<code>acq:sony:f65:cameraProcessDiscriminationCode</code>	unsigned short
Rotary Shutter Mode	<code>acq:sony:f65:rotaryShutterMode</code>	unsigned char
Image Orientation	<code>acq:sony:f65:imageOrientation</code>	unsigned char
Raw Black Code Value	<code>acq:sony:f65:rawBlackCodeValue</code>	unsigned short
Raw Gray Code Value	<code>acq:sony:f65:rawGrayCodeValue</code>	unsigned short
Raw White Code Value	<code>acq:sony:f65:rawWhiteCodeValue</code>	unsigned short

Sony F65 Camera Metadata Set Item Name	Sony-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
Monitoring Base Curve	<code>acq:sony:f65:monitoringBaseCurve</code>	string
Monitoring Base Color	<code>acq:sony:f65:monitoringBaseColor</code>	string
Gamma for Look	<code>acq:sony:f65:gammaforLook</code>	string
Color for Look	<code>acq:sony:f65:colorforLook</code>	string
Pre-CDL Transform	<code>acq:sony:f65:preCdlTransform</code>	string
Post-CDL Transform	<code>acq:sony:f65:postCdlTransform</code>	string
Look Process Baked	<code>acq:sony:f65:lookProcessBaked</code>	unsigned char
Monitoring Characteristics	<code>acq:sony:f65:monitoringCharacteristics</code>	string
Monitoring Color Primaries	<code>acq:sony:f65:monitoringColorPrimaries</code>	string
Monitoring Coding Equations	<code>acq:sony:f65:monitoringCodingEquations</code>	string
Monitoring Descriptions	<code>acq:sony:f65:monitoringDescriptions</code>	string
LED Wall Color Temperature	<code>acq:sony:f65:ledWallColorTemperature</code>	unsigned short
Light Blend Ratio with LED Wall	<code>acq:sony:f65:lightBlendRatioWithLedWall</code>	unsigned char

3.4.3 Camera Posture Metadata Set

3.4.3.1 Sony Private Acquisition Metadata Native Notation

When storing information obtained from each item of the Camera Posture Metadata Set as Sony-specific OpenEXR Attributes, the definitions in the table below SHALL be followed.

However, where normalization to vendor-neutral Attributes defined in the Base Specification, or OpenEXR Standard Attributes, they SHOULD be stored as such.

Camera Posture Metadata Set Item Name	Sony-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
UDAM Set Identifier	<code>acq:sony:camerapos:udamSetIdentifier</code>	string
Camera Relative Position X	<code>acq:sony:camerapos:cameraRelativePositionX</code>	int
Camera Relative Position Y	<code>acq:sony:camerapos:cameraRelativePositionY</code>	int
Camera Relative Position Z	<code>acq:sony:camerapos:cameraRelativePositionZ</code>	int
Camera Pan Angle	<code>acq:sony:camerapos:cameraPanAngle</code>	int
Camera Tilt Angle	<code>acq:sony:camerapos:cameraTiltAngle</code>	int
Camera Roll Angle	<code>acq:sony:camerapos:cameraRollAngle</code>	int
Positioning Data Status	<code>acq:sony:camerapos:positioningDataStatus</code>	unsigned char
Camera Mount Angle	<code>acq:sony:camerapos:cameraMountAngle</code>	unsigned char
Mirror Flip	<code>acq:sony:camerapos:mirrorFlip</code>	unsigned char
Camera System Movability	<code>acq:sony:camerapos:cameraSystemMovability</code>	unsigned char

3.4.3.2 Normalization to OpenEXR Standard Attributes

Among the items in the Camera Posture Metadata Set, those listed in the table below can be stored as the corresponding OpenEXR Standard Attributes, and therefore these SHOULD take precedence.

Camera Posture Metadata Set Item Name	Sony-Specific OpenEXR Attribute Name	OpenEXR Attribute Type	Note
Camera Relative Position X	worldToCamera	M44f	Only with documented CRS conversion
Camera Relative Position Y			
Camera Relative Position Z			
Camera Pan Angle			
Camera Tilt Angle			
Camera Roll Angle			

3.5 Cooke Protocol Lens Attributes

Properties obtained through the Cooke Protocol MAY be stored as Cooke Protocol-specific OpenEXR Attributes in accordance with the table below.

Where possible, these properties of information SHOULD be stored as vendor-neutral lens attributes specified in the Base Specification, or OpenEXR Standard Attributes.

Cooke Protocol Property Name	Cooke Protocol-Specific OpenEXR Attribute Name	OpenEXR Attribute Type
New Start-up Command (NN command response)	acq:cooke:responseNN	string
Retrieve Lens plus Inertial Tracking Data (Kdi command response)	acq:cooke:responseKdi	string
Retrieve one set of Packed Binary Data (Kd command response)	acq:cooke:responseKd	string
Zeiss Extended command (Zd/Zd1 command response)	acq:cooke:responseZd	string
Focus Distance	acq:cooke:focusDistance	unsigned int
Aperture Value	acq:cooke:apertureValue	unsigned short
Aperture Ring T Stop Position	acq:cooke:apertureRingTStopPosition	unsigned short
Zoom	acq:cooke:zoom	unsigned short
Hyperfocal Setting	acq:cooke:hyperfocalSetting	unsigned int
Near Focus Distance	acq:cooke:nearFocusDistance	unsigned int
Far Focus Distance	acq:cooke:farFocusDistance	unsigned int
Horizontal Field of View	acq:cooke:horizontalFieldofView	unsigned short
Entrance Pupil Position	acq:cooke:entrancePupilPosition	unsigned short
Normalized Zoom Value	acq:cooke:normalizedZoomValue	unsigned short
Lens Serial Number	acq:cooke:lensSerialNumber	string

Editor's Note:
The shading or distortion data may need to be parsed further, down to the coefficient level; however, OpenEXR does not provide a convenient way to store arrays.

4. USD Mapping

USD mapping SHALL follow the Base Specification.

The extensions defined in this document do not define independent USD mapping rules except for source-to- `acq:*` normalization and fallback priority.

Recommended USD authoring path:

```
source metadata
↓
acq:<specific>:* specific attributes
↓
acq:* vendor-neutral normalized attributes
↓
OpenEXR Standard Attributes
↓
USD Stage Metadata / UsdGeom.Camera / customData
```

Examples:

```
RDD18 Shutter Speed (Time)
↓
acq:rdd18:camera:shutterSpeedTime
↓
acq:camera:exposureTime
↓
expTime
↓
UsdGeom.Camera.shutter:open / shutter:close
```

```
RDD18 Focus Position From Image Plane
↓
acq:rdd18:lens:focusPositionFromImagePlane
↓
acq:lens:focusDistance
↓
focus
↓
UsdGeom.Camera.focusDistance
```

```
Cooke Real-time Focus Distance
↓
acq:cooke:focusDistance
↓
acq:lens:focusDistance, fallback only
↓
focus
↓
UsdGeom.Camera.focusDistance
```

Sony Camera Posture



acq:sony:camerapos:*



acq:posture:*



worldToCamera, only when CRS conversion is documented



xformOp:transform

Attributes consumed by USD Stage Metadata or `UsdGeom.Camera` SHALL NOT be duplicated into Camera Prim `customData` using the same key.

If source preservation is required, consumed source values MAY be stored under:

`sourceAttributes:<originalKey>`

Examples:

```
sourceAttributes:acq:rdd18:camera:shutterSpeedTime = "1001/60000"  
sourceAttributes:acq:sony:camera:udamSetIdentifier = "20500007-f0c0-1181-9669-08004678031c"  
sourceAttributes:acq:cooke:focusDistance = 3.0
```

